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RAIN SENSOR

ES-RAIN-02

RS485 Modbus RTU

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1. Product Overview

The Rain and Snow Sensor is a device primarily used to detect the presence of rain or snow in natural environments. This sensor employs an AC impedance measurement method, ensuring a long electrode lifespan and preventing oxidation issues. It is widely applicable for qualitative measurement of rain and snow in various settings such as environmental monitoring, greenhouses, aquaculture, construction, and buildings, offering safety, reliability, aesthetic design, and ease of installation.

The sensor utilizes AC impedance measurement, which effectively avoids electrode oxidation and electrolysis, significantly extending its operational life. It provides accurate rain and snow measurement results with a near-zero false alarm rate. An optional heating function is available, which automatically activates when low temperatures are detected to accelerate snow and ice removal, thereby increasing the measurement speed.

2. Technical Specifications

Parameter	**Value**
Power Supply	10–30V DC
Storage Temperature	-40℃ to +80℃
Operating Temperature	-40℃ to +60℃
Default Modbus Address	0X01
Heating Activation Temp	<15℃ (default)
Relay Output Load Capacity	250VAC 1A / 30VDC 1A

Normal Operating Power	0.4W
Heating Typical Power	3.4W
Output Signal	RS485, Relay
Supported Function Codes	03, 06
Max Heating Temp	40℃ (default)
Parameter Configuration	Software setup

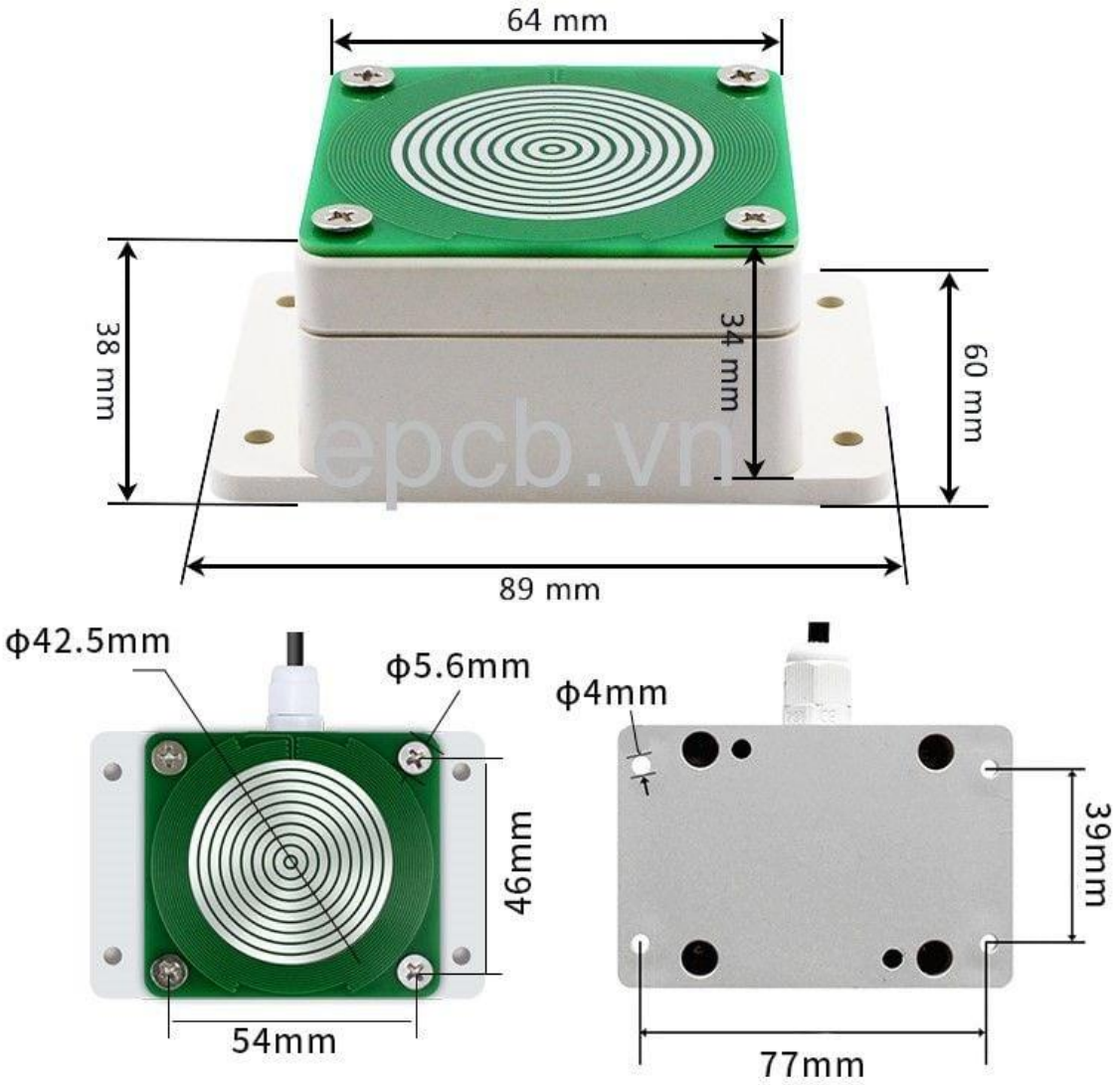
3. Product Discription

3.1. Product photo





3.2. Dimension



3.3. Wiring

Wire Color	Description
Brown	Power supply (V+)
Black	Ground (V-)

Yellow	RS485 A line (Data +)
Blue	RS485 B line (Data -)
Green/White	Relay output

4. Installation method

Install the device with a 15° tilt from the horizontal surface.



Angle

5. Communication

5.1. Basic Communication Parameters

Parameter	**Value**
Address Code	8-bit binary
Data Bits	8 bits
Parity Check	None
Stop Bits	1 bit
Error Check	CRC (Cyclic Redundancy Check)
Baud Rate	Configurable: 2400, 4800, 9600 bps; Default: 4800 bps

5.2. Data Frame Format Definition

Modbus-RTU protocol is used. Frame format:

- Start Frame: ≥ 4 bytes idle time
- Device Address: 1 byte
- Function Code: 1 byte
- Data Area: N bytes
- CRC Check: 16-bit CRC
- End Frame: ≥ 4 bytes idle time

Address: Unique identifier of transmitter on the Modbus network (default: 0x01)

Function Code: This device uses only 0x03 (Read Holding Registers)

Data Area: Specific transmission data (16-bit, high byte first)

CRC: 2-byte verification code

Master Query Frame

Field	**Address**	**Function**	**Start Addr**	**Register Length**	**CRC Low**	**CRC High**
Bytes	1	1	2	2	1	1

Slave Response Frame

Field	**Address**	**Function**	**Byte Count**	**Data 1**	**Data 2**	**...**	**CRC Low**	**CRC High**
Bytes	1	1	1	2	2	...	1	1

5.3. Register address

Register Address	PLC or Group Address	Description	Operation	Default Value	Range
0000H	40001	Real-time rain and snow status	Read only	0	0 or 1
0030H	40049	Heating upper limit temperature	R/W	35°C	0–70°C
0031H	40050	Heating lower limit temperature	R/W	15°C	-30–70°C
0032H	40051	Heating temperature hysteresis	R/W	5°C	0–70°C
0033H	40052	Alarm recovery delay time	R/W	1s	0–60000s
0034H	40053	Sensitivity setting	R/W	800	500–3500

5.4. Communication Protocol Example and Explanation

Reading rain/snow status of device at address 0x01

Request Frame:

Address	Function Code	Start Address	Register Count	CRC Low Byte	CRC High Byte
0x01	0x03	0x00 0x00	0x00 0x01	0x84	0x0A

Response Frame (Normal Rain/Snow Status):

Address	Function Code	Byte Count	Data	CRC Low Byte	CRC High Byte
0x01	0x03	0x02	0x00 0x00	0xB8	0x44

Rain/Snow Status Definition:

Code	Status
0x00	Normal
0x01	Alarm

Read alarm recovery delay at device address 0x01

Request Frame:

Address	Function	Start Address	Data Length	CRC Low	CRC High
0x01	0x03	0x00 0x33	0x00 0x01	0x74	0x05

Response Frame (Example: Recovery delay is 1s):

Address	Function	Byte Count	Data Field	CRC Low	CRC High
0x01	0x03	0x02	0x00 0x01	0x79	0x84

Set alarm recovery delay at device address 0x01 (Example: 10s)

Request Frame:

Address	Function	Start Address	Data Value	CRC Low	CRC High
0x01	0x06	0x00 0x33	0x00 0x0A	0xF9	0xC2

Response Frame (Echo):

Address	Function	Start Address	Data Value	CRC Low	CRC High
0x01	0x06	0x00 0x33	0x00 0x0A	0xF9	0xC2

Note:

When the set delay is 10s, the device will enter alarm mode if snowfall persists beyond 10 seconds and trigger output. If snowfall ends before 10 seconds, the device remains in normal status and does not enter alarm mode.

Default Value: 1s

Range: 0–60000s

Read Current Sensitivity of Device Address 0x01

Request Frame:

Address	Function	Start Address	Data Length	CRC Low	CRC High
0x01	0x03	0x00 0x34	0x00 0x01	0xC5	0xC4

Response Frame (Example: Sensitivity = 800):

Address	Function	Byte Count	Data Field	CRC Low	CRC High
0x01	0x03	0x02	0x03 0x20	0xB9	0x06

Set Current Sensitivity to 1500 at Address 0x01 (Example)

Request Frame:

Address	Function	Start Address	Data	CRC Low	CRC High
0x01	0x06	0x00 0x34	0x05 0xDC	0xCA	0x0D

Response Frame (Echo):

Address	Function	Start Address	Data	CRC Low	CRC High
0x01	0x06	0x00 0x34	0x05 0xDC	0xCA	0x0D

Sensitivity Setting Description

The sensitivity value determines the sensor's reaction to snowfall. A lower value reduces detection sensitivity, while a higher value increases it. Overly high sensitivity may lead to false alarms. It is recommended to use default settings unless otherwise necessary.

- Default Value: 800
- Range: 500–3500